

JOINS AND SET OPERATIONS

```
[mysql> SELECT * FROM class_info;
```

id	address
1	DELHI
2	MUMBAI
3	CHENNAI

```
3 rows in set (0.000 sec)
```

```
mysql> SELECT class.name, class_info.address  
-> FROM class  
-> CROSS JOIN class_info;
```

name	address
alex	DELHI
adam	DELHI
abhi	DELHI
alex	MUMBAI
adam	MUMBAI
abhi	MUMBAI
alex	CHENNAI
adam	CHENNAI
abhi	CHENNAI

```
9 rows in set (0.000 sec)
```

```
[mysql> SELECT * FROM class;
```

id	name
1	abhi
2	adam
4	alex

```
3 rows in set (0.000 sec)
```

```
[mysql> TRUNCATE TABLE class;
```

```
Query OK, 0 rows affected (0.010 sec)
```

```
mysql> INSERT INTO class VALUES
```

```
-> (1,'abhi'),  
-> (2,'adam'),  
-> (3,'alex'),  
-> (4,'anu');
```

```
Query OK, 4 rows affected (0.002 sec)
```

```
Records: 4 Duplicates: 0 Warnings: 0
```

```
[mysql> SELECT * FROM class;
```

id	name
1	abhi
2	adam
3	alex
4	anu

```
4 rows in set (0.000 sec)
```

```

mysql> SELECT * FROM class_info;
+-----+-----+
| id | address |
+-----+-----+
| 1 | DELHI |
| 2 | MUMBAI |
| 3 | CHENNAI |
+-----+-----+
3 rows in set (0.000 sec)

mysql> SELECT class.name, class_info.address
-> FROM class
-> INNER JOIN class_info
[ -> ON class.id = class_info.id;
+-----+-----+
| name | address |
+-----+-----+
| abhi | DELHI |
| adam | MUMBAI |
| alex | CHENNAI |
+-----+-----+
3 rows in set (0.000 sec)

mysql> SELECT class.name,
-> class_info.address
-> FROM class
-> INNER JOIN class_info
[ -> ON class.id = class_info.id;
+-----+-----+
| name | address |
+-----+-----+
| abhi | DELHI |
| adam | MUMBAI |
| alex | CHENNAI |
+-----+-----+
3 rows in set (0.001 sec)

mysql> SELECT *
-> FROM class
[ -> NATURAL JOIN class_info;
+-----+-----+
| id | name | address |
+-----+-----+
| 1 | abhi | DELHI |
| 2 | adam | MUMBAI |
| 3 | alex | CHENNAI |
+-----+-----+
3 rows in set (0.000 sec)

mysql> INSERT INTO class VALUES
[ -> (5,'ashish');
Query OK, 1 row affected (0.002 sec)

mysql> INSERT INTO class_info VALUES
-> (7,'NOIDA'),
[ -> (8,'PANIPAT');
Query OK, 2 rows affected (0.001 sec)
Records: 2 Duplicates: 0 Warnings: 0

```

```
[mysql> SELECT * FROM class;
```

id	name
1	abhi
2	adam
3	alex
4	anu
5	ashish

```
5 rows in set (0.000 sec)
```

```
[mysql> SELECT * FROM class_info;
```

id	address
1	DELHI
2	MUMBAI
3	CHENNAI
7	NOIDA
8	PANIPAT

```
5 rows in set (0.000 sec)
```

```
mysql> SELECT class.name,  
-> class_info.address  
-> FROM class  
-> LEFT OUTER JOIN class_info  
-> ON class.id = class_info.id;
```

name	address
abhi	DELHI
adam	MUMBAI
alex	CHENNAI
anu	NULL
ashish	NULL

```
5 rows in set (0.001 sec)
```

```
mysql> SELECT *  
-> FROM class  
-> LEFT OUTER JOIN class_info  
-> ON class.id = class_info.id;
```

id	name	id	address
1	abhi	1	DELHI
2	adam	2	MUMBAI
3	alex	3	CHENNAI
4	anu	NULL	NULL
5	ashish	NULL	NULL

```
5 rows in set (0.001 sec)
```

```
mysql> SELECT *
-> FROM class
-> LEFT JOIN class_info
-> ON class.id = class_info.id
[ -> WHERE class_info.id IS NULL;
```

id	name	id	address
4	anu	NULL	NULL
5	ashish	NULL	NULL

2 rows in set (0.001 sec)

```
mysql> SELECT class.id, class.name
-> FROM class
-> LEFT JOIN class_info
-> ON class.id = class_info.id
[ -> WHERE class_info.id IS NULL;
```

id	name
4	anu
5	ashish

2 rows in set (0.001 sec)

```
mysql> SELECT *
-> FROM class
-> RIGHT OUTER JOIN class_info
[ -> ON class.id = class_info.id;
```

id	name	id	address
1	abhi	1	DELHI
2	adam	2	MUMBAI
3	alex	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

5 rows in set (0.001 sec)

```
mysql> SELECT class_info.id, class_info.address
-> FROM class
-> RIGHT JOIN class_info
-> ON class.id = class_info.id
[ -> WHERE class.id IS NULL;
```

id	address
7	NOIDA
8	PANIPAT

2 rows in set (0.001 sec)


```
mysql> SELECT *
-> FROM class
-> LEFT JOIN class_info
-> ON class.id = class_info.id
->
-> UNION
->
-> SELECT *
-> FROM class
-> RIGHT JOIN class_info
-> ON class.id = class_info.id;
```

id	name	id	address
1	abhi	1	DELHI
2	adam	2	MUMBAI
3	alex	3	CHENNAI
4	anu	NULL	NULL
5	ashish	NULL	NULL
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

7 rows in set (0.003 sec)

```
mysql> SELECT class.id, class.name, class_info.id, class_info.address
-> FROM class
-> LEFT JOIN class_info
-> ON class.id = class_info.id
-> WHERE class_info.id IS NULL
->
-> UNION
->
-> SELECT class.id, class.name, class_info.id, class_info.address
-> FROM class
-> RIGHT JOIN class_info
-> ON class.id = class_info.id
-> WHERE class.id IS NULL;
```

id	name	id	address
4	anu	NULL	NULL
5	ashish	NULL	NULL
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

4 rows in set (0.001 sec)

```
mysql> CREATE TABLE first_table (
-> id INT,
-> name VARCHAR(30)
-> );
Query OK, 0 rows affected (0.009 sec)
```

```
mysql> CREATE TABLE second_table (
-> id INT,
-> name VARCHAR(30)
-> );
Query OK, 0 rows affected (0.014 sec)
```

```
mysql> INSERT INTO first_table VALUES
    -> (1, 'abhi'),
[    -> (2, 'adam');
Query OK, 2 rows affected (0.003 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO second_table VALUES
    -> (2, 'adam'),
[    -> (3, 'chester');
Query OK, 2 rows affected (0.003 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
[mysql> SELECT * FROM first_table;
```

id	name
1	abhi
2	adam

2 rows in set (0.000 sec)

```
[mysql> SELECT * FROM second_table;
```

id	name
2	adam
3	chester

2 rows in set (0.000 sec)

```
mysql> SELECT * FROM first_table
    -> UNION
[    -> SELECT * FROM second_table;
```

id	name
1	abhi
2	adam
3	chester

3 rows in set (0.001 sec)

```
mysql> SELECT * FROM first_table
    -> UNION ALL
[    -> SELECT * FROM second_table;
```

id	name
1	abhi
2	adam
2	adam
3	chester

4 rows in set (0.000 sec)

```
mysql> SELECT name FROM first_table
-> UNION
[ -> SELECT name FROM second_table;
```

name
abhi
adam
chester

3 rows in set (0.001 sec)

```
mysql> SELECT name FROM first_table
-> UNION ALL
[ -> SELECT name FROM second_table;
```

name
abhi
adam
adam
chester

4 rows in set (0.001 sec)

```
mysql> SELECT * FROM first_table
-> INTERSECT
[ -> SELECT * FROM second_table;
```

id	name
2	adam

1 row in set (0.001 sec)

```
[mysql> SELECT name FROM first_table;
```

name
abhi
adam

2 rows in set (0.000 sec)

```
[mysql> SELECT name FROM second_table;
```

name
adam
chester

2 rows in set (0.000 sec)


```
[mysql> SELECT name FROM second_table;
```

```
+-----+
```

```
| name |
```

```
+-----+
```

```
| adam |
```

```
| chester |
```

```
+-----+
```

```
2 rows in set (0.000 sec)
```

```
mysql> SELECT name FROM first_table
```

```
-> INTERSECT
```

```
[ -> SELECT name FROM second_table;
```

```
+-----+
```

```
| name |
```

```
+-----+
```

```
| adam |
```

```
+-----+
```

```
1 row in set (0.001 sec)
```

```
mysql> SELECT f.*
```

```
-> FROM first_table f
```

```
-> LEFT JOIN second_table s
```

```
-> ON f.id = s.id
```

```
[ -> WHERE s.id IS NULL;
```

```
+-----+-----+
```

```
| id | name |
```

```
+-----+-----+
```

```
| 1 | abhi |
```

```
+-----+-----+
```

```
1 row in set (0.001 sec)
```

```
mysql> SELECT f.name
```

```
-> FROM first_table f
```

```
-> LEFT JOIN second_table s
```

```
-> ON f.name = s.name
```

```
[ -> WHERE s.name IS NULL;
```

```
+-----+
```

```
| name |
```

```
+-----+
```

```
| abhi |
```

```
+-----+
```

```
1 row in set (0.001 sec)
```

```
[mysql> exit
```

```
Bye
```